10.5 H(JW) from a differential eq

Or from a difference equation.

We are considering linear time invariant systems, so if we find the response for one thing, we know it for everything.

Take, for example

$$\ddot{y}(t) + \alpha\dot{y}(t) + \beta y(t) = \gamma x(t)$$

Let $x = e^{j\omega t}$
We know that

$$y = H(j\omega)e^{j\omega t}$$

We can plug in, divide by $e^{j\omega t}$(because its never zero), and solve for $H(j\omega)$. This is because $e^{j\omega t}$ has the eigenvalue property.

$$\gamma(e^{j\omega t}) = -\omega^2 H(j\omega)(e^{j\omega t}) + \alpha j\omega H(j\omega)(e^{j\omega t}) + \beta H(j\omega)(e^{j\omega t})$$

becomes

$$\begin{aligned}\gamma &= -\omega^2 H(j\omega) + \alpha j\omega H(j\omega) + \beta \\ \gamma &= H(j\omega)(-\omega^2 + \alpha j\omega + \beta) \\ H(j\omega) &= \frac{\gamma}{-\omega^2 + \alpha j\omega + \beta}\end{aligned}$$